

- 3.2** (a) How many moles of C atoms are in 1 mol of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)?
 (b) How many C atoms are in 2 mol of sucrose?

3.8 Calculate the molar mass of each of the following:

- (a) $\text{Sr}(\text{OH})_2$ (b) N_2O_3 (c) NaClO_3 (d) Cr_2O_3

3.12 Calculate each of the following quantities:

- (a) Mass in grams of 0.68 mol of KMnO_4
 (b) Moles of O atoms in 8.18 g of $\text{Ba}(\text{NO}_3)_2$
 (c) Number of O atoms in 7.3×10^{-3} g of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

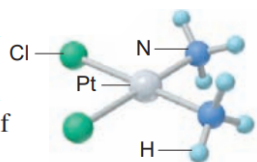
3.16 Calculate each of the following quantities:

- (a) Mass in grams of 8.35 mol of copper(I) carbonate
 (b) Mass in grams of 4.04×10^{20} molecules of dinitrogen pentoxide
 (c) Number of moles and formula units in 78.9 g of sodium perchlorate
 (d) Number of sodium ions, perchlorate ions, Cl atoms, and O atoms in the mass of compound in part (c)

3.18 Calculate each of the following:

- (a) Mass % of H in ammonium bicarbonate
 (b) Mass % of O in sodium dihydrogen phosphate heptahydrate

3.23 Cisplatin (*right*), or Platinol, is used in the treatment of certain cancers. Calculate (a) the moles of compound in 285.3 g of cisplatin; (b) the number of hydrogen atoms in 0.98 mol of cisplatin.



3.28 The mineral galena is composed of lead(II) sulfide and has an average density of 7.46 g/cm^3 . (a) How many moles of lead(II) sulfide are in 1.00 ft^3 of galena? (b) How many lead atoms are in 1.00 dm^3 of galena?

3.33 What is the empirical formula and empirical formula mass for each of the following compounds?

- (a) C_2H_4 (b) $\text{C}_2\text{H}_6\text{O}_2$ (c) N_2O_5 (d) $\text{Ba}_3(\text{PO}_4)_2$ (e) Te_4I_{16}

3.39 An oxide of nitrogen contains 30.45 mass % N. (a) What is the empirical formula of the oxide? (b) If the molar mass is $90 \pm 5 \text{ g/mol}$, what is the molecular formula?

3.44 Cortisol ($M = 362.47 \text{ g/mol}$), one of the major steroid hormones, is a key factor in the synthesis of protein. Its profound effect on the reduction of inflammation explains its use in the treatment of rheumatoid arthritis. Cortisol is 69.6% C, 8.34% H, and 22.1% O by mass. What is its molecular formula?

3.53 Write balanced equations for each of the following by inserting the correct coefficients in the blanks:

- (a) $\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow \text{SO}_3(\text{g})$
 (b) $\text{Sc}_2\text{O}_3(\text{s}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \text{Sc}(\text{OH})_3(\text{s})$
 (c) $\text{H}_3\text{PO}_4(\text{aq}) + \text{NaOH}(\text{aq}) \longrightarrow \text{Na}_2\text{HPO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 (d) $\text{C}_6\text{H}_{10}\text{O}_5(\text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$

3.55 Convert the following into balanced equations:

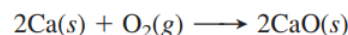
- (a) When gallium metal is heated in oxygen gas, it melts and forms solid gallium(III) oxide.
 (b) Liquid hexane burns in oxygen gas to form carbon dioxide gas and water vapor.
 (c) When solutions of calcium chloride and sodium phosphate are mixed, solid calcium phosphate forms and sodium chloride remains in solution.

3.63 Chlorine gas can be made in the laboratory by the reaction of hydrochloric acid and manganese(IV) oxide:



When 1.82 mol of HCl reacts with excess MnO_2 , (a) how many moles of Cl_2 form? (b) How many grams of Cl_2 form?

3.73 Many metals react with oxygen gas to form the metal oxide. For example, calcium reacts as follows:



You wish to calculate the mass of calcium oxide that can be prepared from 4.20 g of Ca and 2.80 g of O_2 .

- (a) How many moles of CaO can be produced from the given mass of Ca?
 (b) How many moles of CaO can be produced from the given mass of O_2 ?
 (c) Which is the limiting reactant?
 (d) How many grams of CaO can be produced?

3.79 Aluminum nitrite and ammonium chloride react to form aluminum chloride, nitrogen, and water. What mass of each substance is present after 72.5 g of aluminum nitrite and 58.6 g of ammonium chloride react completely?

3.95 Are the following instructions for diluting a 10.0 M solution to a 1.00 M solution correct: "Take 100.0 mL of the 10.0 M solution and add 900.0 mL water"? Explain.

3.96 Calculate each of the following quantities:

- (a) Grams of solute in 185.8 mL of 0.267 M calcium acetate
 (b) Molarity of 500. mL of solution containing 21.1 g of potassium iodide
 (c) Moles of solute in 145.6 L of 0.850 M sodium cyanide

3.100 Calculate each of the following quantities:

- (a) Molarity of a solution prepared by diluting 37.00 mL of 0.250 *M* potassium chloride to 150.00 mL
- (b) Molarity of a solution prepared by diluting 25.71 mL of 0.0706 *M* ammonium sulfate to 500.00 mL
- (c) Molarity of sodium ion in a solution made by mixing 3.58 mL of 0.348 *M* sodium chloride with 500. mL of 6.81×10^{-2} *M* sodium sulfate (assume volumes are additive)

3.102 A sample of concentrated nitric acid has a density of 1.41 g/mL and contains 70.0% HNO₃ by mass.

- (a) What mass of HNO₃ is present per liter of solution?
- (b) What is the molarity of the solution?

3.114 Narceine is a narcotic in opium that crystallizes from solution as a hydrate that contains 10.8 mass % water and has a molar mass of 499.52 g/mol. Determine *x* in narceine·*x*H₂O.

3.134 Assuming that the volumes are additive, what is the concentration of KBr in a solution prepared by mixing 0.200 L of 0.053 *M* KBr with 0.550 L of 0.078 *M* KBr?

3.161 Cocaine (C₁₇H₂₁O₄N) is a natural substance found in coca leaves, which have been used for centuries as a local anesthetic and stimulant. Illegal cocaine arrives in the United States either as the pure compound or as the hydrochloride salt (C₁₇H₂₁O₄NHCl). At 25°C, the salt is very soluble in water (2.50 kg/L), but cocaine is much less so (1.70 g/L). (a) What is the maximum amount (in g) of the salt that can dissolve in 50.0 mL of water? (b) If the solution in part (a) is treated with NaOH, the salt is converted to cocaine. How much additional water (in L) is needed to dissolve it?